Homework. Chistiakova Mariia. ME317

№ 1.

for bond i, defining the default indicator:

$$y_i = \begin{cases} 1, & \text{if bond defaults} \\ 0, & \text{otherwise} \end{cases}$$

then $y_i \sim \text{Bernoulli}(0.03)$ and $y_1, \dots, y_N$ are independent

Step 1:

the current price is 98

if the bond does not default $\Rightarrow$ its value in one year is 104 $\Rightarrow$

$\Rightarrow L_i = 98 - 104 = -6 \Rightarrow$ there is a profit of 6

if bond defaults $\Rightarrow$ its future value is 0 $\Rightarrow L_i = 98 - 0 = 98$

therefore we get $L_i = \begin{cases} 98, & y_i = 1 \\ -6, & y_i = 0 \end{cases}$

or we can write $L_i = 98 y_i - 6(1 - y_i) = 104 y_i - 6$

Step 2:

the portfolio contains one unit of every bond $\Rightarrow L = \sum_{i=1}^{N} L_i = 104 \sum_{i=1}^{N} y_i - 6N$

let $D = \sum_{i=1}^{N} y_i$ be the total number of defaults

as defaults are independent $\Rightarrow D \sim \text{Bin}(N, 0.03)$

hence the exact loss is $L = 104 D - 6N$

so exact VaR may be written as $\text{VaR}_{0.95}(L) = 104\, q_{0.95}(\text{Bin}(N, 0.03)) - 6N$

Step 3:

for a binomial random variable $E[D] = 0.03N$ and $\text{Var}(D) = N(0.03)(0.97)$

for sufficiently large N, $D \approx N(0.03N,\ 0.03(0.97)N)$

as $L = 104D - 6N \Rightarrow E[L] = 104(0.03N) - 6N = -2.88N$

$\text{Var}(L) = 104^2 (0.03)(0.97) N$

for a normally distributed loss $\text{VaR}_\alpha(L) = \mu_L + \sigma_L \Phi^{-1}(\alpha)$

using $\Phi^{-1}(0.95) \approx 1.645$ we obtain:

$$VaR_{0,95}(L) \approx -2.88\,N + 1.645\,(104)\sqrt{0.03\,(0.97)\,N}$$

thus $VaR_{0.95}(L) \approx -2.88\,N + 29.2\sqrt{N}$

a negative VaR is possible here, it means that at the 95% quantile level the portfolio still makes a profit because the promised payoff 104 is higher than current price 98.

№ 2.

□ : suppose $X \sim N(0, a^2)$, $Y \sim N(0, b^2)$, $a \geq 0$, $b \geq 0$

X and Y are independent

for a normal random variable with mean zero : $VaR_\alpha(X) = a\,\Phi^{-1}(\alpha)$

$VaR_\alpha(Y) = b\,\Phi^{-1}(\alpha)$

as X and Y are independent $X + Y \sim N(0, a^2 + b^2)$

therefore $VaR_\alpha(X+Y) = \sqrt{a^2+b^2}\;\Phi^{-1}(\alpha)$

for the usual risk-management levels $\alpha > 0.5$: $\Phi^{-1}(\alpha) > 0$

also $\sqrt{a^2+b^2} \leq a + b$ as $a^2 + b^2 \leq a^2 + 2ab + b^2 = (a+b)^2$

multiplying by positive number $\Phi^{-1}(\alpha)$ we obtain $VaR_\alpha(X+Y) \leq (a+b)\,\Phi^{-1}(\alpha)$

but $(a+b)\,\Phi^{-1}(\alpha) = VaR_\alpha(X) + VaR_\alpha(Y)$

hence $VaR_\alpha(X+Y) \leq VaR_\alpha(X) + VaR_\alpha(Y)$

therefore VaR satisfies subadditivity for independent normal risk ∎

№ 3.

we have $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}\right)$

initial portfolio value is $V_0 = 100$

for a stock portfolio with weights ω_1, ω_2 : $L^\Delta = -V_0(\omega_1 X_1 + \omega_2 X_2)$

① Strategy A: invest equally in both stocks

$\omega_{A1} = \omega_{A2} = \frac{1}{2}$

so $L_A^{\Delta} = -100\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = -50X_1 - 50X_2$

Strategy B: invest everything in stock 1

$\omega_{B1} = 1, \ \omega_{B2} = 0$

so $L_B^{\Delta} = -100\left(1\cdot X_1 + 0\cdot X_2\right) = -100X_1$

② a linear combination of a multivariate normal vector is normally distributed with variance $\omega' \Sigma \omega$

$(X_1, X_2) \sim N_2(\mu, \Sigma)$

$\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \Rightarrow Var(X_1) = 1$

$Var(X_2) = 2$

$Cov(X_1, X_2) = 0.5$

Strategy A.

the portfolio return is $\frac{1}{2}X_1 + \frac{1}{2}X_2$

$Var\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{4}Var X_1 + \frac{1}{4}Var X_2 + 2\cdot\frac{1}{2}\cdot\frac{1}{2}Cov(X_1, X_2) = \frac{1}{4} + \frac{1}{4} + 2\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} =$

$= \frac{3}{4}$

so $Var(L_A^{\Delta}) = 100^2 \cdot \frac{3}{4}$

$\sigma_A = 100\sqrt{\frac{3}{4}} = 50\sqrt{3}$

$VaR_{0.99}(L_A^{\Delta}) = 2.33\left(50\sqrt{3}\right)$

Strategy B:

as $L_B^{\Delta} = -100X_1$ and $Var(X_1) = 1 \Rightarrow \sigma = 100$

so $VaR_{0.99}(L_B^{\Delta}) = 2.33\cdot(100)$

therefore the ratio is $\frac{VaR_{0.99}(L_A^{\Delta})}{VaR_{0.99}(L_B^{\Delta})} = \frac{2.33\left(50\sqrt{3}\right)}{2.33\cdot(100)} = \frac{\sqrt{3}}{2} \approx 0.865$

thus the equally weighted portfolio has approximately 86.5% of the VaR of the fully concentrated portfolio, this is the diversification effect

③ let $U = X_{1,day\,1}$ be log return of stock 1 during the first day

let $W = X_{2,day\,2}$ be log return of stock 2 during the second day

as returns on different days are independent $U \sim N(0,1)$, $W \sim N(0,1)$ and

U and W are independent

therefore $R = U + W \sim N(0, 2)$

Exact loss :

after the first day, the portfolio value is $V_1 = 100e^{U}$

all of this amount is invested in stock 2 so after the second day :

$V_2 = 100 e^{U} e^{W} = 100 e^{U+W} = 100 e^{R}$

hence exact two-day loss : $L = 100 - V_2 = 100(1 - e^{R})$

large losses occur when R is very negative

the 99% quantile of L therefore corresponds to 1% quantile of R

since $R \sim N(0,2) \Rightarrow q_{0.01} R = -2.33\sqrt{2}$ ($e^{-2.33\sqrt{3}}$ would be used if the two returns had cov = 0.5, as if they occured on the same day, however we are told that returns on different days are independent)

consequently $VaR_{0.99}(L) = 100(1 - e^{-2.33\sqrt{2}}) \approx 100(1 - 0.037) \approx 96.3$

Linearized loss :

using $e^{R} \approx 1 + R$ we get $L = 100(1 - e^{R}) \approx 100(1-(1+R)) = -100R$

thus $L^{\Delta} = -100(U + W)$

since $U + W \sim N(0, 2)$: $L^{\Delta} \sim N(0, 100^2 \cdot 2)$

its standard deviation is $\sigma_{L^{\Delta}} = 100\sqrt{2}$

therefore $VaR_{0.99}(L^{\Delta}) = 2.33(100\sqrt{2}) \approx 329.5$

the linearized VaR is much larger than exact VaR because the given standard deviation of the log returns is extremely large, the first-log order approximation works best for small risk-factor changes and short, relatively calm periods

Great work !